

Kiddies Town ECD & Academy

Vercel + Neon Deployment Guide

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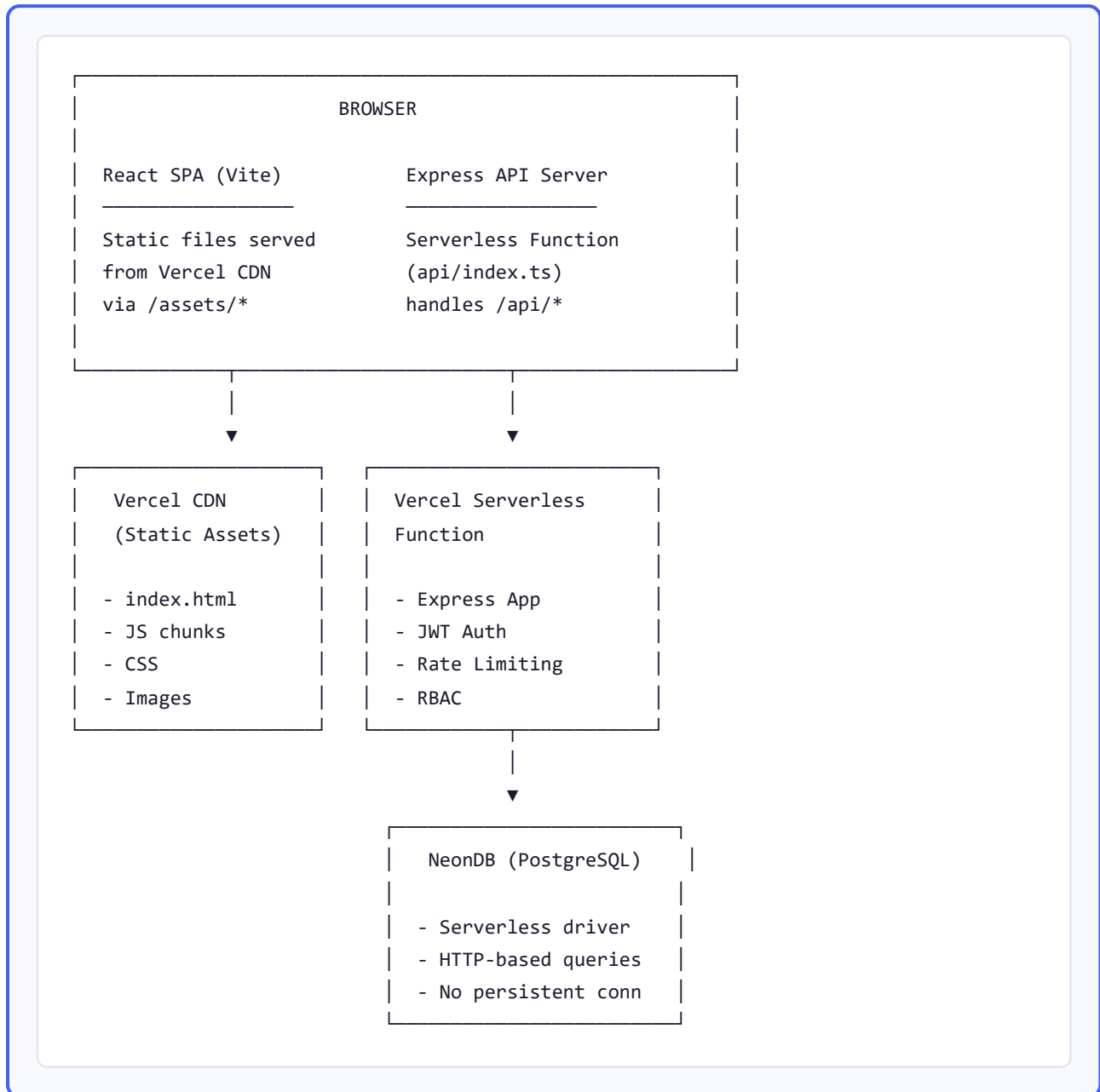
Platform: Vercel (Frontend + API) + NeonDB (PostgreSQL)

Stack: React 19, TypeScript, Vite, Express 4, NeonDB

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1. Architecture Overview



Request Flow

1. Browser requests `https://your-app.vercel.app/` → Vercel serves `dist/index.html`
2. Browser requests `https://your-app.vercel.app/api/data/learners` → Vercel rewrites to serverless function
3. Serverless function runs Express app → queries NeonDB via HTTP → returns JSON
4. React Query caches the response client-side

2. Prerequisites

Requirement	Version	Notes
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Node.js	18+	Recommended: 20 LTS
npm	9+	
Git	2.30+	
GitHub account	—	For repository hosting
Vercel account	—	Free tier sufficient for start
Neon account	—	Free tier: 0.5 GB storage

3. Environment Variables

Required (set in Vercel Dashboard)

Variable	Value	How to Generate
<code>DATABASE_URL</code>	<code>postgresql:// ...</code>	Neon dashboard → Connection Details
<code>JWT_SECRET</code>	64+ char random string	<code>openssl rand -hex 64</code>
<code>JWT_REFRESH_SECRET</code>	64+ char random string	<code>openssl rand -hex 64</code>
<code>NODE_ENV</code>	<code>production</code>	Static value
<code>CORS_ORIGIN</code>	<code>https://your-app.vercel.app</code>	Your Vercel domain

Optional

Variable	Default	Notes
<code>GEMINI_API_KEY</code>	—	Google AI PDF generation
<code>APP_URL</code>	—	Used for email links
<code>PORT</code>	<code>3000</code>	Ignored on Vercel

Setting Environment Variables on Vercel

```
# Via Vercel CLI
vercel env add DATABASE_URL production
vercel env add JWT_SECRET production
vercel env add JWT_REFRESH_SECRET production
vercel env add NODE_ENV production
vercel env add CORS_ORIGIN production

# Or via Vercel Dashboard:
# Project Settings → Environment Variables → Add
```

4. Repository Setup

4.1 Initialize Git (if not already done)

```
cd "path/to/kiddies-town-portal"
git init
git add -A
git commit -m "feat: initial deployment configuration"
```

4.2 Create .gitignore Entries

```
# Environment
.env
.env.local
.env.*.local

# Build
dist/
.tmp/

# Data
data_store.json

# Dependencies
node_modules/

# IDE
.vscode/
.idea/
```

4.3 Push to GitHub

```
git remote add origin https://github.com/your-username/kiddies-town-portal.git
git branch -M main
git push -u origin main
```

5. Vercel Deployment

5.1 Import Project

1. Go to vercel.com/new
2. Click **"Import Git Repository"**
3. Select your GitHub repository
4. Configure:
 - **Framework Preset:** Other
 - **Build Command:** `npm run build`
 - **Output Directory:** `dist`
 - **Install Command:** `npm install`
5. Click **Deploy**

5.2 Configure Environment Variables

After deployment, go to **Project Settings** → **Environment Variables** and add:

```
DATABASE_URL = postgresql://neondb_owner:xxxx@ep-xxxxx.us-east-1.aws.neon.tech/neondb
JWT_SECRET = <your-generated-secret>
JWT_REFRESH_SECRET = <your-generated-secret>
NODE_ENV = production
CORS_ORIGIN = https://your-project.vercel.app
```

5.3 Redeploy

After setting environment variables, trigger a redeployment:

```
vercel --prod
```

Or click **"Redeploy"** in the Vercel dashboard.

5.4 Custom Domain (Optional)

1. Go to **Project Settings** → **Domains**
2. Add your custom domain (e.g., `kiddiestown.co.za`)
3. Update `CORS_ORIGIN` to include your custom domain

4. Redeploy

6. Neon Database Setup

6.1 Create Neon Project

1. Go to neon.tech
2. Sign up / log in
3. Create a new project
4. Note the connection string (you'll need it for `DATABASE_URL`)

6.2 Connection String Format

```
postgresql://neondb_owner:password@ep-xxxxx.us-east-1.aws.neon.tech/neondb?sslmode=r
```

⚠ Use the **Pooled** connection string for serverless (Vercel). The pooler handles connection multiplexing for concurrent serverless functions.

6.3 Schema Bootstrap

On first deployment, the app automatically:

1. Creates all database tables (12 tables)
2. Creates PostgreSQL enum types
3. Seeds initial data (learners, themes, events, etc.)

✓ This only happens once — subsequent deployments skip seeding if data exists.

6.4 Production Safety

The `bootstrapSchema()` function is gated for production:

- **Development:** Drops and recreates tables (for schema iteration)
- **Production:** Only creates missing tables/indexes (preserves existing data)

7. Post-Deployment Verification

7.1 Health Check

```
curl https://your-app.vercel.app/api/health
# Expected: {"status":"healthy","timestamp":" ... ","version":"2.0.0"}
```

7.2 Login Test

```
curl -X POST https://your-app.vercel.app/api/v1/auth/login \
  -H "Content-Type: application/json" \
  -d '{"email":"admin@kiddiestown.co.za","password":"admin"}'
# Expected: {"success":true,"token":" ... ","refreshToken":" ... ","user":{" ... }}
```

7.3 Data Fetch Test

```
# First login to get token, then:
curl https://your-app.vercel.app/api/v1/data \
  -H "Authorization: Bearer <token>"
# Expected: JSON with learners, themes, events, etc.
```

7.4 Frontend Test

1. Open `https://your-app.vercel.app` in browser
2. Verify landing page loads
3. Click **Login** → enter credentials
4. Verify dashboard loads with data
5. Test CRUD operations (add/edit/delete)

8. Troubleshooting

Issue: "Invalid environment variables" on deploy

Cause: Missing or invalid `JWT_SECRET` / `JWT_REFRESH_SECRET`

Fix: Ensure both secrets are at least 32 characters. Generate with:

```
node -e "console.log(require('crypto').randomBytes(64).toString('hex'))"
```

Issue: "Neon query failed, falling back to local store"

Cause: `DATABASE_URL` not set or invalid

Fix:

1. Verify `DATABASE_URL` is set in Vercel dashboard
2. Check the Neon connection string format
3. Ensure Neon project is active (not paused)

Issue: CORS errors in browser console

Cause: `CORS_ORIGIN` doesn't match your Vercel domain

Fix: Update `CORS_ORIGIN` to `https://your-exact-domain.vercel.app`

Issue: "DROP TABLE" errors in production logs

Cause: Bootstrap dropping tables on cold start

Fix: Already fixed — production mode only creates missing tables. Ensure `NODE_ENV=production` is set.

Issue: 502 Bad Gateway on API routes

Cause: Serverless function timeout or cold start

Fix:

- 1. Check function logs in Vercel dashboard
- 2. Increase `maxDuration` in `vercel.json` (current: 30s)
- 3. Check Neon connection health

Issue: Static assets 404

Cause: `outputDirectory` misconfigured

Fix: Ensure `vercel.json` has `"outputDirectory": "dist"` and `npm run build` outputs to `dist/`

9. Files Modified for Deployment

File	Changes
<code>vercel.json</code>	New — Vercel configuration (rewrites, headers, functions)
<code>api/index.ts</code>	New — Serverless function entry point
<code>server/db/bootstrap.ts</code>	Gated table drops for production
<code>server/config/database.ts</code>	Disabled file I/O in serverless mode
<code>server/app.ts</code>	CORS auto-allows Vercel preview deployments
<code>.env.example</code>	Updated with security guidance

Appendices

Appendix A: Vercel CLI Commands

```
# Install Vercel CLI
npm i -g vercel

# Login
vercel login

# Deploy (preview)
vercel

# Deploy (production)
vercel --prod

# Pull environment variables
vercel env pull .env.local

# View function logs
vercel logs

# Remove project
vercel rm kiddies-town-portal
```

Appendix B: Neon CLI Commands

```
# Install Neon CLI
npm i -g neonctl

# List projects
neon projects list

# Create connection string
neon connection-string <project-id>
```

Appendix C: Cost Estimate

Service	Free Tier	Paid (Starting)
Vercel	100 GB bandwidth, 1000 build min	\$20/mo (Pro)
Neon	0.5 GB storage, 24/7 compute	\$19/mo (Launch)
Total	\$0/mo	\$39/mo

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